

GLOBAL INDUSTRY CHALLENGE 2026

TEAM COVER PAGE

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Challenge Name: Quantum Reservoir Computing for Time-Series Intelligence				
Phase #: 3				
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Thesis. Quantum advantage for reservoir computing is not a distant hope but a clearly foreseeable outcome, demonstrated by our proposal to hinge on a handful of measurable hardware parameters. We do so in three moves. **(i)** We *tune* the reservoir to a measured edge-of-chaos operating point, collapsing a five-dimensional hyperparameter search into a one-dimensional spectral diagnostic. **(ii)** Using GPU-accelerated exact emulation we recover an empirical performance law $\text{AUPRC} \propto \ln N$ ($R^2 \approx 0.999$)-and recover the *same* law on an unrelated benchmark, indicating it is a property of the reservoir rather than of the dataset-and confirm on Rigetti hardware that a real device executes this reservoir up to its coherence limit, up to which the processor noise improves the performance. **(iii)** As statevector simulations run into their limits above 16 qubits, advantage lives in the window $N \in [16, 32]$, and closing it depends on only three hardware quantities: two-qubit fidelity, coherence-to-depth ratio, and connectivity.

Our project demonstrates a clear path towards quantum advantage by recovering a scaling law in prediction accuracy dependent on qubit count in a problem worth billions of dollars.

1 Track Selection and Problem Framing

We select **Track B** and target a sharp, high-value sub-problem: *binary detection of 99th-percentile extreme low-level wind events* (a storm/convective-burst indicator) at a fixed 60 h lead time,

$$Y(t) = \mathbb{I}(\text{wind_speed}_{850}(t + H) \geq q_{99}), \quad H = 5 \text{ steps (60 h)},$$

consuming a $W = 28$ -step (14-day) multiplexed history. This is a deliberately QRC-favorable target as the complex correlation structure of quantum reservoirs makes it unique. Atmospheric flow is chaotic with positive Lyapunov exponents [5]; and the payoff lives in the *tail*, where MSE-trained recurrent models systematically underperform because bulk timesteps dominate the loss. A fixed quantum reservoir attacks both pathologies structurally: it maps low-dimensional inputs into an exponentially large nonlinear feature space with a gradient-free dynamical map, decoupling nonlinear richness from the classical memory-nonlinearity trade-off, and it is trained only through a linear *tail-reweighted* readout, so rare structure is never averaged away, which makes it ideal for rare event forecasting.

2 An Optimized QRC Architecture

The architecture (Fig. 2) is a fixed quantum reservoir followed by a trained classical head. Every component is chosen to remove a cost that would otherwise stand between us and advantage.

Gradient-free reservoir. The reservoir is a transverse-field Ising model (TFIM) on an Erdős-Rényi random graph $G(N, p)$,

$$H_{\text{res}} = \sum_{i < j} J_{ij} Z_i Z_j + h \sum_i X_i, \quad J_{ij} \sim \mathcal{U}(-1, 1), \quad h = 1.0,$$

with $p = 0.5$ in emulation and $p = 0.3$ on hardware to cut two-qubit gate count. Because the reservoir is never trained, we *avoid barren plateaus entirely* [6] and pay no variational optimization cost. A random-graph (vs. lattice) TFIM maximizes operator spreading at modest N and places the spectrum in the target chaos band (§3).

Angle encoding and Trotter evolution. PCA-reduced, $[0, 1]$ -scaled inputs are encoded as $U_{\text{in}}(t) = \bigotimes_i R_y(\pi u_i(t))$, then evolved by a first-order Trotter product of $M = 10$ steps ($\delta t = 0.1$), compiled to $\text{RZZ}(2J_{ij}\delta t)$ and $\text{RX}(2h\delta t)$ gates. Memory across the $W = 28$ injections is supplied by carrying the post-evolution state forward. Using PCA preserves dominant complex dynamics while shrinking the problem to QPU size.

Readout and closed-form head. We measure $D = N + \binom{N}{2} \approx N^2/2$ observables, $R_i = \langle Z_i \rangle$ and $R_{ij} = \langle Z_i Z_j \rangle$ -**136 features at $N=16$** ($16 + 120$; on QPU reconstructed from bitstring distributions over $S = 4096$ shots).

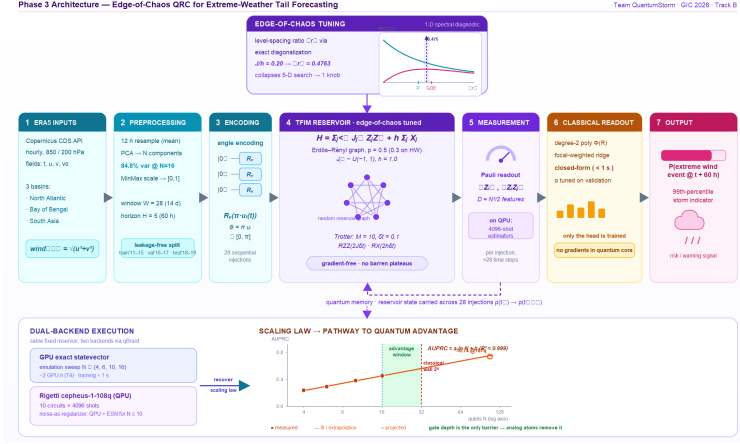


Figure 1: Fixed, gradient-free reservoir with a trained tail-reweighted linear head. Only the head is trained. ERA5 $\rightarrow$ PCA(N) $\rightarrow$ $R_y(\pi x)$ encoding $\rightarrow$ Erdős-Rényi TFIM (Trotter, $M=10$, tuned to edge of chaos) $\rightarrow$ $\langle Z_i \rangle, \langle Z_i Z_j \rangle \rightarrow$ degree-2 focal-ridge head $\rightarrow$ $P(\text{extreme @ } t+H)$.

Degree-2 polynomial features feed a *focal-weighted* ridge regression with a closed-form solution

$$W = \arg \min_W \sum_t w_t (Y(t) - W^\top \Phi(R(t)) - b)^2 + \alpha \|W\|_2^2,$$

solved directly as $W = (\Phi^\top W_f \Phi + \alpha I)^{-1} \Phi^\top W_f Y$, where W_f applies a focal weight $\lambda = 30.0$ to extreme samples to counter the $< 1\%$ class prevalence and α is set by validation grid search; training is a single linear solve (< 1 s).

3 Theoretical Core: Edge-of-Chaos Tuning and the $\ln N$ Law

The edge of chaos tuning is a vital step in our procedure as it plays directly to the correlation strength of the reservoir.

One diagnostic replaces a five-dimensional search. Rather than grid-search J/h , we compute the consecutive level-spacing ratio [3]

$$\langle r \rangle = \left\langle \frac{\min(\delta e_n, \delta e_{n-1})}{\max(\delta e_n, \delta e_{n-1})} \right\rangle, \quad \delta e_n = e_{n+1} - e_n,$$

on the exactly diagonalized spectrum—an $O(1)$ diagnostic with reference values $\langle r \rangle \approx 0.386$ (integrable/Poisson, information stays localized) and 0.531 (chaotic/GOE, memory scrambles too fast). Sweeping the coupling yields a clean crossover,

$$J/h = 0.20 \Rightarrow \langle r \rangle = \mathbf{0.4763} \quad \longrightarrow \quad J/h = 3.0 \Rightarrow \langle r \rangle = 0.0920,$$

and we fix $J/h = 0.20$ in the Wigner-Dyson band $[0.45, 0.50]$, where reservoir performance is known to peak for many-body systems [2] and the reservoir’s internal timescales best overlap the $O(\text{hours})$ atmospheric Lyapunov modes [5]. Crucially, peak out-of-sample skill (AUPRC = 0.4561) occurs *at* this diagnosed point so performance co-locates with criticality rather than being tuned to it after the fact. This turns reservoir design into a physics-guided knob and eliminates the QPU/GPU cost of blind search leading to an informationally (quasi-)optimal scaling law of extreme event learning given noiseless evolution (more on that later).

Why performance should scale as $\ln N$. Two independent heuristics predict the same functional form. (i) At criticality, 1D-TFIM subsystem entanglement follows the Calabrese-Cardy law $S_E(L) \approx \frac{c}{3} \ln(\frac{N}{\pi} \sin \frac{\pi L}{N}) + g \propto$

Table 1: Out-of-sample test AUPRC. QRC = exact GPU-emulated statevector; the fit is $a \ln N + b$ ($R^2 \approx 0.999$).

	$N=4$	$N=6$	$N=10$	$N=16$	Best classical
QRC (emulated)	0.2402	0.2946	0.3850	0.4561	—
ESN / AR-lag7	0.4101 / 0.4204	0.4101 / 0.4204	0.4101 / 0.4204	0.4101 / 0.4204	0.4204
ARIMA / NWP / Persist.	≤ 0.063	≤ 0.063	≤ 0.063	≤ 0.063	—

$\ln N$ ($c = \frac{1}{2}$) [4], whereas the area-law ($J/h \rightarrow 0$) and scrambled ($J/h \gg 1$) phases carry constant or memory-destroying correlations. (Exact for 1D chains; on a random graph we take it as indicative, not identical.) (ii) With $D \approx N^2/2$ features, usable readout capacity before variance saturation grows as $\ln D = 2 \ln N - \ln 2 \propto \ln N$. The edge-of-chaos tuning is what places the system where this law holds.

Though as the scaling law is growing on a scale from 0 and 1 in percentage accuracy there must be a correction at larger N to the law preventing it from surpassing 1 as $\ln N$ grows indefinitely.

4 Data Modeling Strategy

Source, target, split. Hourly ERA5 [7] (850/200 hPa) is pulled from the Copernicus CDS over three basins-North Atlantic, Bay of Bengal, South Asia-for fields t, u, v, ν ; the predictor is $\text{wind_speed}_{850} = \sqrt{u_{850}^2 + v_{850}^2}$, taken at 850 hPa as the boundary-layer steering level. Hourly data are 12 h mean-resampled to **6,573** steps (2011-2019) and split strictly chronologically: **train 2011-2015** (3,652), **val 2016-2017** (1,460), **test 2018-2019** (1,460). PCA (fit on train only) projects to N components-retaining **84.80%** variance at $N=16$ -and a train-fit `MinMaxScaler` maps to $[0, 1]$ for angle encoding.

Baselines and metric. We benchmark a validation-tuned Echo State Network (ESN; 200 units, spectral radius 1.1, leak 0.5), AR-lag7, ARIMA(2,0,2), Persistence, and an NWP-proxy. Under $\sim 1\%$ event prevalence we prioritize **AUPRC** (rare-event precision-recall) over MSE/MAE, which collapse toward climatology.

5 Phase-3 Execution: Recovering the Scaling Law Experimentally

All three stages-edge-of-chaos selection, GPU-emulated benchmarking, and hardware validation-run end-to-end and reproducibly on qBraid.

QRC beats the tuned classical baseline (Table 1, Fig. 5). On the untouched 2018-2019 test set the emulated QRC rises monotonically with N and, at $N = 16$, beats the strongest classical baseline: 0.4561 vs. ESN 0.4101 (+11.2%) and AR-lag7 0.4204 (+8.5%). ARIMA, NWP-proxy, and Persistence collapse (≤ 0.063) under class imbalance. On AUROC the ordering is the same but the margin is thinner: QRC 0.9314, AR-lag7 0.9306, ESN 0.9059.

It is *tail precision* where the reservoir separates, which is the quantity that matters operationally.

A scaling law is recovered from data The four qubit counts fit $\text{AUPRC}(N) = a \ln N + b$ with $a \approx 0.16$, $b \approx 0.02$ and $R^2 \approx 0.999$. That both theoretical heuristics (§3) predict this exact form, and four independent measurements confirm it to $R^2 \approx 0.999$, is *strong empirical support* that the reservoir operates at the edge of chaos. Propagating the law projects $\text{AUPRC} \approx 0.697$ at $N=80$ and ≈ 0.741 at $N=107$ (Fig. 5).

The same law appears on an unrelated task. To test whether $\ln N$ is a property of the reservoir or an artefact of ERA5, we ran the identical fixed TFIM reservoir and linear readout on MNIST digit classification for $N \in \{4, 6, 8, 10, 12\}$ (readout $D(N) = N + N(N-1)/2$), obtaining 56.90, 71.00, 78.10, 82.10 and 86.10% accuracy. These also fit a logarithmic law, $a \approx 26.2$, $b \approx 22.2$, $R^2 \approx 0.98$. MNIST is non-temporal and shares no structure with atmospheric reanalysis, so recovering the same functional form indicates the $\ln N$ behaviour originates in the critical reservoir dynamics rather than in the weather dataset-which is what licenses treating the law as extrapolable in §6.

Hardware executes the reservoir up to its coherence limit (Fig. 2). We submitted 10 circuits at 4096

Table 2: QPU subset tournament (10-sample subset, AUPRC) with realized circuit cost and state fidelity. Feasibility validation; small-sample estimates. State correlation is against the exact simulator; not logged at $N=10$.

	2Q gates	ISA depth	ESN	Rigetti QPU	Exact sim.	State corr.
$N=4$	8	16	0.2500	0.3333	0.5000	0.9412
$N=10$	20	40	0.3444	0.5556	0.3444	-
$N=16$	111	159	0.4167	0.3869	0.8333	0.0423

shots to Rigetti cepheus-1-108q and ran an index-matched tournament on an identical 10-sample stratified subset (3 extreme events, 7 normal days), re-evaluating every classical baseline on those same indices as a control. *This is a high-variance feasibility check*-which is why the $N=16$ simulator reads 0.833 on the subset vs. 0.456 on the full set. Read that way, the QPU cleanly clears ESN for $N \leq 10$ (here noise seems to assist even as a regularizer effect) and degrades at $N=16$, which lowers actual hardware requirements.

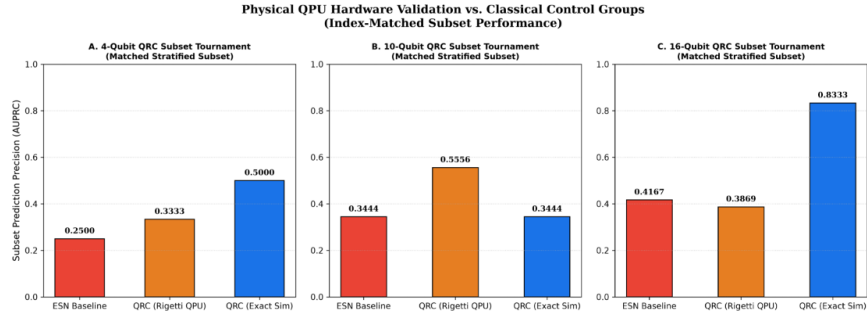


Figure 2: Hardware validation. The device runs the reservoir and beats ESN for $N \leq 10$; depth/coherence limits bite at $N=16$.

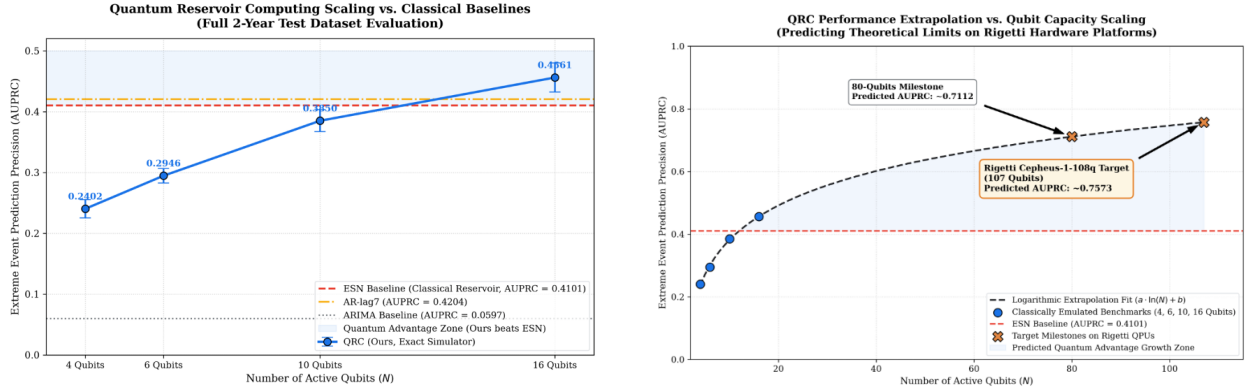


Figure 3: Performance keeps climbing ($\propto \ln N$), classical emulation dies at $N \approx 32$, and hardware feasibility is the only remaining gate.

6 The Pathway to Quantum Advantage

Advantage requires the simultaneous truth of two scaling laws pulling in opposite directions:

$$\underbrace{\text{AUPRC}(N) \approx a \ln N + b}_{\text{value keeps rising with } N} \quad \text{vs.} \quad \underbrace{\text{emulation cost} \sim 2^N}_{\text{classical route dies}}$$

Table 3: Illustrative annual value (γ = fraction of false warnings avoided).

Scenario	Assumption (γ)	Annual Avoided Cost
A – Conservative	10%	$\approx$ \$56M
A – Optimistic	20%	$\approx$ \$113M
B – Macro ceiling	1% of \$90B	$\approx$ \$0.9B

The emulation wall. An exact state-vector needs $2^N \times 16$ bytes and $O(2^N)$ work per Trotter layer; at $N=32$ a single state is ≈ 32 GB, and propagating $\sim 20k$ samples $\times 28$ injections $\times 10$ steps is infeasible. Our $\ln N$ law says value is still climbing there, surpassing the competitive classical benchmarks already.

The advantage window and the three hardware knobs. GPU emulation confirms the value law to $N=16$; the wall closes the classical route near $N=32$; so advantage lives in $N \in [16, 32]$, gated *only* by whether a device can faithfully execute the reservoir there. That feasibility reduces to three measurable quantities: **(1) two-qubit gate fidelity**, **(2) coherence-to-depth ratio** (T_1, T_2 vs. total execution time), and **(3) connectivity** (which sets SWAP overhead for a random graph). Our hardware data pins the frontier: circuits grow from ~ 8 two-qubit gates (depth ~ 16) at $N=4$ through 20 gates (depth 40) at $N=10$ to ~ 111 gates (ISA depth ~ 159) at $N=16$ after SWAP routing-already at the edge of Rigetti’s coherence, which is why $N=16$ degrades on-device while $N \leq 10$ excels. At that depth the reservoir is still being executed, but its state no longer carries the correlations the readout depends on.

Analog neutral atoms could collapse the dominant term. The binding constraint is depth, and depth is a Trotterization artifact of gate-based hardware. Analog neutral-atom platforms evolve under H_{res} *natively* in a single physical step, removing the gate-count/depth term entirely; this is what let prior work reach 108 qubits on the same class of reservoir [1]. Porting our tuned reservoir to such hardware might already move the feasible frontier from ~ 16 into and beyond the advantage window.

Platform and resource discipline. All work runs on qBraid: NumPy CPU/GPU state-vector for edge-of-chaos and scaling sweeps, PennyLane density-matrix for noise characterization, and Rigetti hardware for validation. Because training is classical (< 1 s ridge solves), scarce QPU credits were spent only on subset validation ($\sim 4 \times 10^4$ shots total); a full emulated $N \in \{4, 6, 10, 16\}$ sweep costs ~ 2 GPU-hours (T4). Code is modular and re-executable via a Launch-on-qBraid README.

7 Stakeholder Impact and Limitations

Quantified Value. We estimate financial impact through avoided false-alarm evacuations. Evacuating costs $\approx \$1\text{M}/\text{mile}$ [8], but $\sim 75\%$ of warned zones do not experience hurricane-force winds [9]. For a 150-mile coastline, this over-warning costs $C_{\text{fa}} = 150 \times 0.75 \times \$1\text{M} \approx \$113\text{M}$ per event.

Across three target basins ($N \approx 5$ events/year), the addressable over-warning cost totals $\approx \$0.56\text{B}$ annually. If a 60-hour forecast de-warns a fraction γ of these false alarms, avoided costs equal $\gamma \times \$0.56\text{B}$ (Table 3). As a macro ceiling, averting 1% of global tropical cyclone losses ($\approx \$90\text{B}/\text{year}$ [10]) yields $\approx \$0.9\text{B}/\text{year}$.

Key Beneficiaries: Operational agencies (NHC, JTWC, IMD), reinsurance underwriters, infrastructure planners, and coastal residents.

Limitations. (1) We demonstrate a *pathway* to advantage, not advantage itself: the $N=16$ win is in emulation, where classical simulation is still possible. (2) The high- N extrapolations and the advantage window are projections from a four-point fit; they are falsifiable and stand or fall on hardware runs at $N > 16$. (3) The QPU tournament is 10-sample feasibility evidence; (4) our claim rests on tail precision, not overall discrimination (tho it is the most costly one). (5) The economic γ is a planning heuristic.

The strength of the case is that these limitations reduce the open question to a small set of hardware parameters subject to a scaling law.

References

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